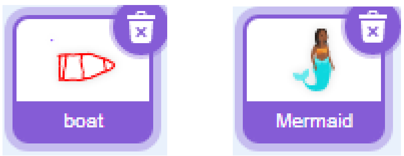
Boat maze!

Based on the instructions https://projects.raspberrypi.org/en/projects/boat-race/

See the finished project at https://scratch.mit.edu/projects/811047590/.

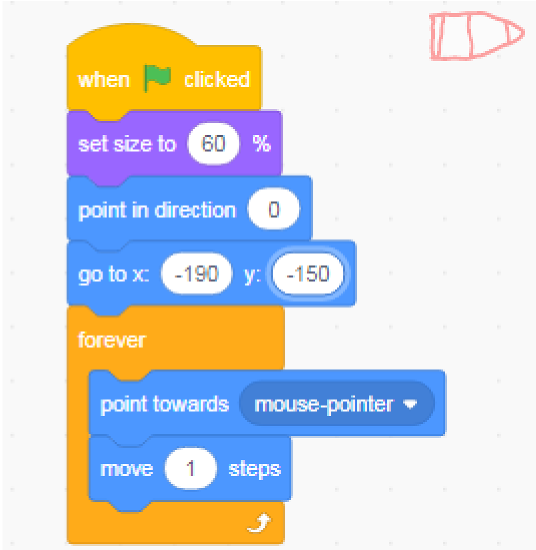
Step 1: Drawing

- Start off by drawing a maze, make sure to leave gaps the boat can get through!
- The colour you use should be the same for all of the jetty (the maze walls).
- If you go wrong, use the Undo arrow  to go back one or more steps.
- You can draw a boat or use a Scratch sprite for your game.



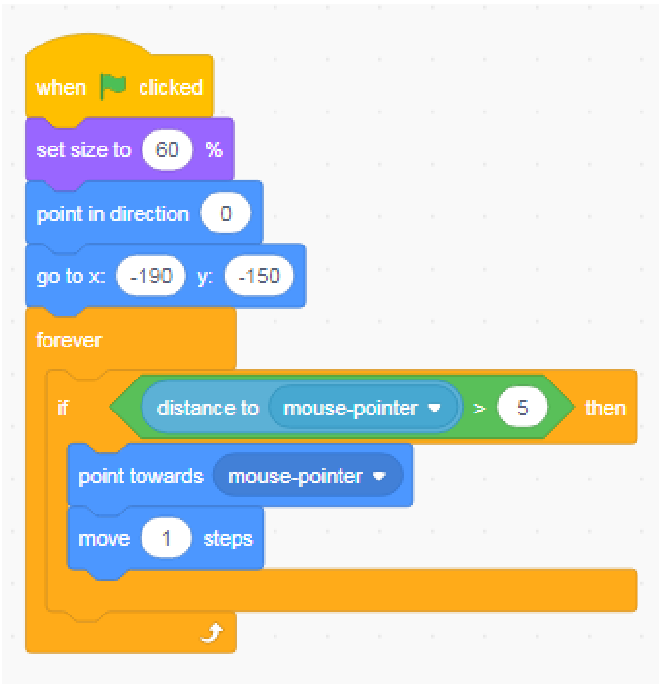
Save your project

Step 2: Move your boat



- Add the code above to your boat sprite, and test it out – does it move?
- You may have different numbers for the size and x-y position
- What happens when your boat catches up with the mouse?

- Add the blocks above to stop your boat glitching, so your code looks like the picture below:



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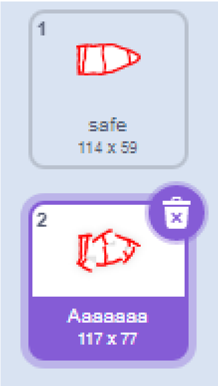
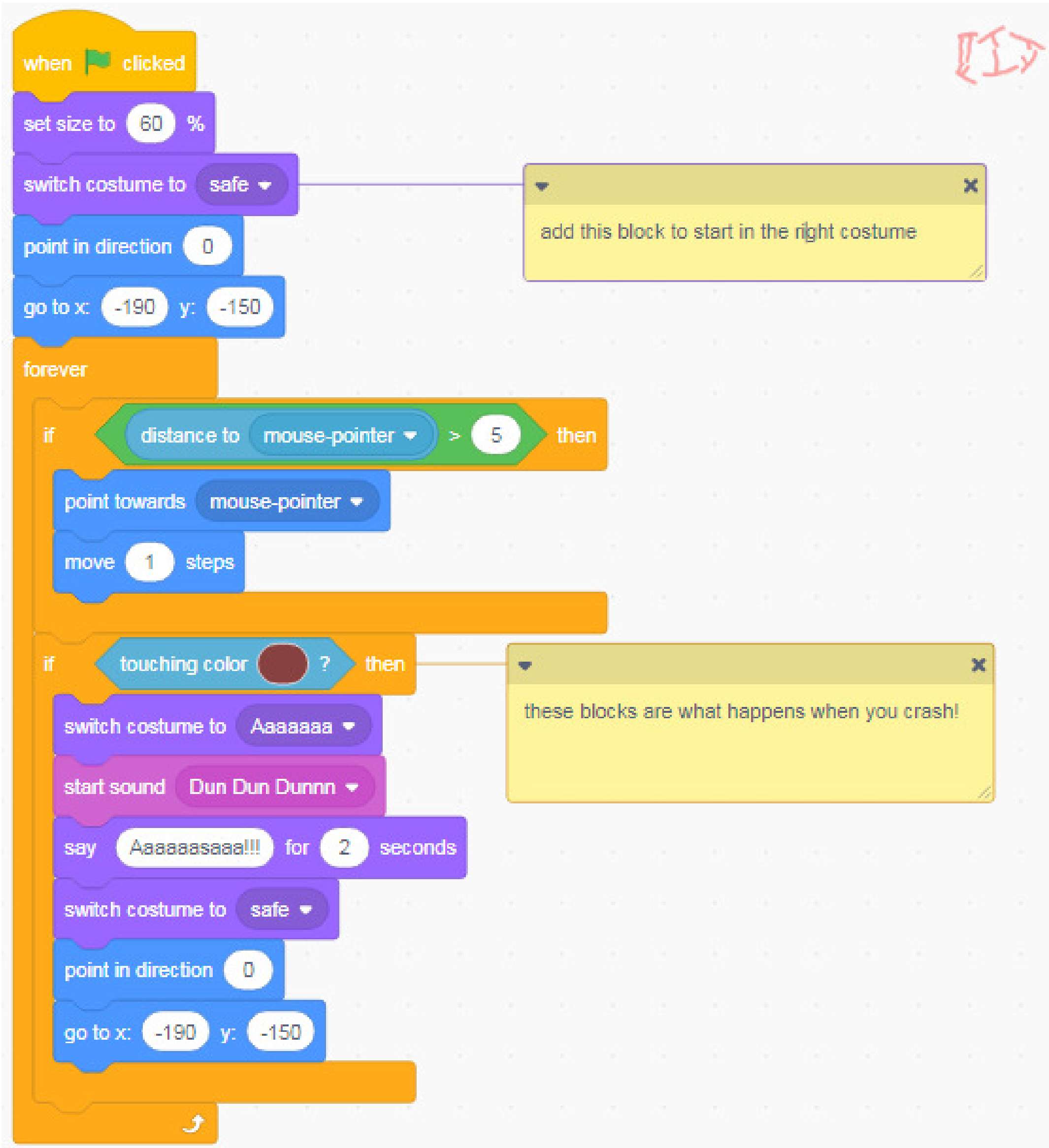
Step 3: Crash!



Add a ‘crashed’ costume to your sprite, by duplicating it and then using the selection tool to move bits of it around.

Next add the blocks below to your code to add crashing to your game:



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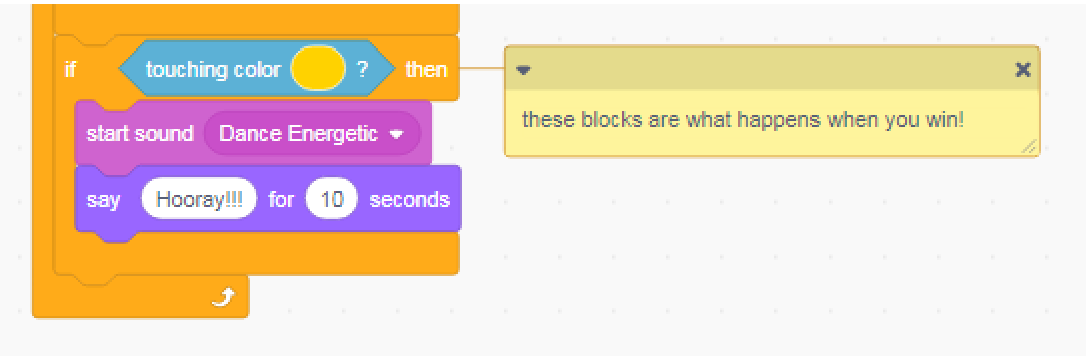
Step 4: Winning!

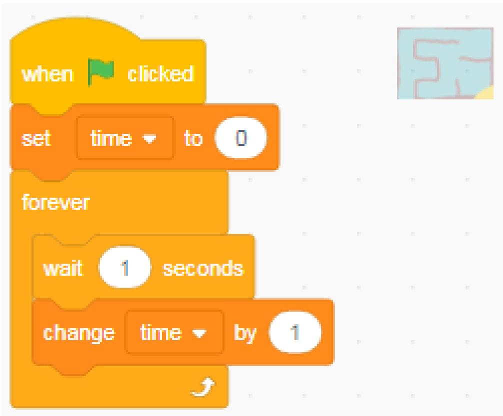
- Add another If statement right at the end of your code, and choose the blocks that will show you have won the game!



Step 5: A timer

Make your game more exiting by adding a timer:

- Make a variable and call it ‘time’
- Put the code below on the backdrop:



Save your project

Step 6: A shark!

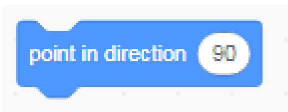
Can you add a moving hazard, like a shark?

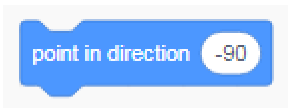
- First draw or choose your sprite:

- Then to work out where it will move, put it in the first position, and choose this block:

- Your numbers will be different, but don't change them! Move the sprite to the second position it will move to, and drag the glide block in again (don't change the numbers!).
- Put the two glide blocks in a forever loop, like this:

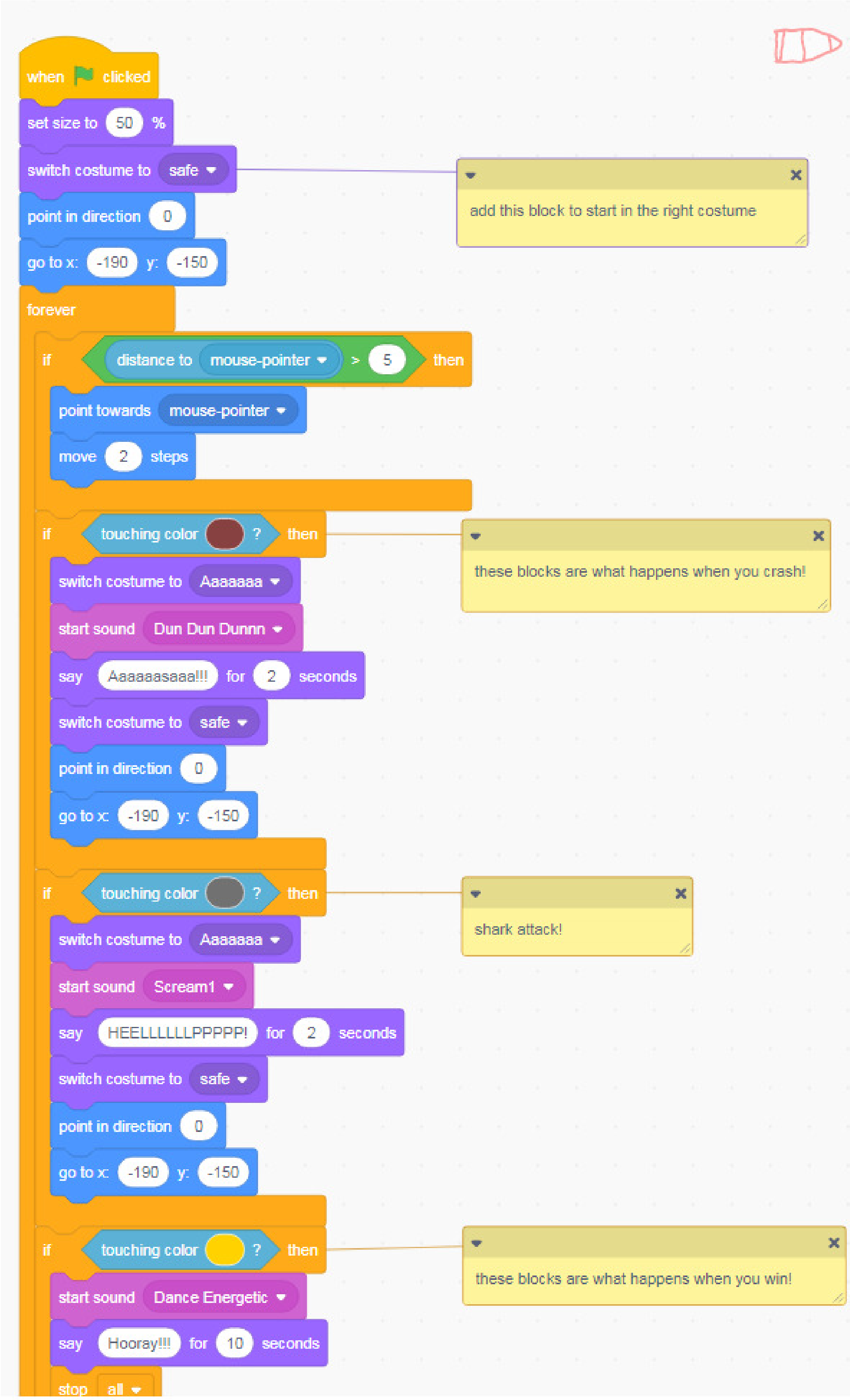
- You might need these blocks too:





- Test it out! You may need to tweak the numbers a bit.
- Once you are happy, add some code to the boat sprite for when you touch the shark – see the next page for the full code. Can you survive the shark?

Save your project



Save your project

Step 7: Speed up and slow down – and a menu!

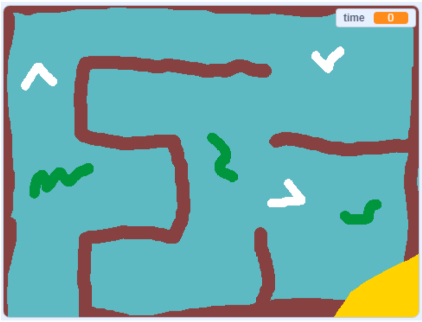
Let’s make the game a bit harder:

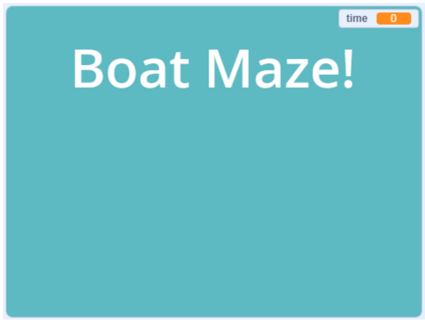
- We can add white waves to give a speed boost
- And patches of seaweed to slow us down

But what if we want to choose between the easy game and the hard game?

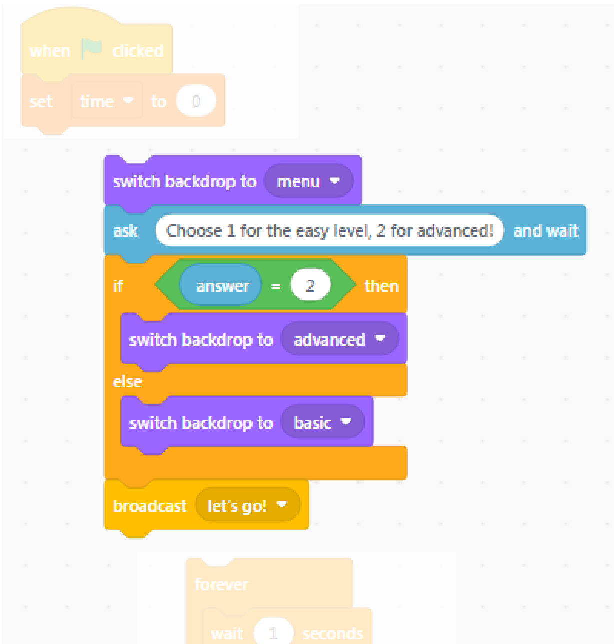
- Let’s add a menu! First, rename your backdrop to ‘basic’:

- Then make a copy, call it ‘advanced’, and draw on some white waves and green seaweed. And finally add a plain backdrop and type the title on it:



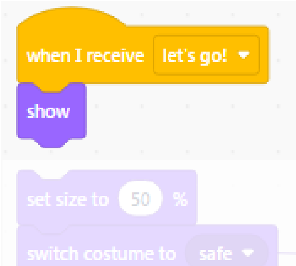


- Add the code below INSIDE the backdrop code you already have (shown paler):



Now let's change our sprite code, so nothing starts until we've chosen a level.

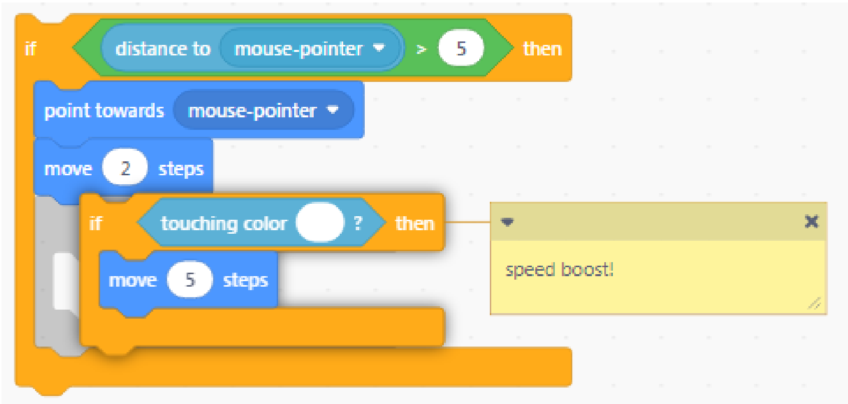
- Break the boat code away from the green flag, and replace with a 'when I receive' message block, and 'show'. Then add 'hide' to the green flag block:



- Make the same change to your shark sprite ('when I receive', 'show', 'hide').
- Test your code – can you choose the different game levels? We haven't coded the waves and seaweed yet, but does everything else work?

Save your project

The code for the speed boost is an 'if' statement that goes INSIDE the 'if' you already have – the one to check for glitching – like this:



- And the code for the seaweed goes OUTSIDE and AFTER the glitching 'if':

- Test your code again – does your boat speed up and slow down?
- Can you win the game on the advanced level?

Save your project